

# Ritabrata Chakraborty

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## EDUCATION

### University of Pennsylvania

*M.S.E. in Robotics, GPA: -/4*

Pennsylvania, USA

Aug '26 – May '28

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

*B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10*

Rajasthan, India

Oct '22 – May '26

## PUBLICATIONS & PEER REVIEWS

### Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

### Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

### Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

### Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

## RESEARCH EXPERIENCE

### Research Intern

Supervisor: [Dr. Avinash Gautam](#), Associate Professor

BITS Pilani

May '25 – Aug '25

- Literature study of vision-language grounded multi-robot coordination and VLMs-based navigation.

### Research Intern

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist

CSIR-CEERI

Jan '24 – May '26

- Monocular vision-based UAV navigation for orchard monitoring using YOLOv11. [🔗](#)
- Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation. [🔗](#)

### Research Intern

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor

MARMoT Lab, NUS

Mar '25 – Dec '25

- Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan). [🔗](#)
- Vision-attention-driven autonomous navigation with semantic understanding, extending [CogniPlan](#) and [ViNT](#).

### Research Assistant

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor

BITS Pilani

Apr '24 – Oct '25

- Physics-based launcher mechanical design and FEM analysis. [🔗](#)
- Computer vision pipeline for 3D trajectory reconstruction and shot classification using [PnLCalib](#).
- Finite element analysis of running track surfaces using hyperelastic material models.

## Research Intern

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor

- Synthetic data generation for bearing fault diagnosis under data scarcity. [🔗](#)
- Physics-informed generative modeling for bearing fault diagnosis under data scarcity, validated on [LiteFormer](#).

**KU Leuven**

Sep '24 – May '26

## Research Assistant

Supervisor: [Dr. Rajneesh Choubisa](#)

- Primal world beliefs study surveying engineering undergraduates on core worldview dimensions. [🔗](#)

**BITS Pilani**

Aug '24 – Dec '24

## Research Intern

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD

- Image-based visual servoing for automated radar control and UAV tracking using YOLOv8 and DeepSORT.
- Impact testing and analysis of metal-composite shields using electromagnetic railguns.

**BARC**

Jun '24 – Jul '24

## INDUSTRY EXPERIENCE

### ML Intern

Supervisors: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer

- Object annotation automation and anomaly detection in [uLabel](#).

**Uber**

Jul '25 – Dec '25

### Subject Matter Expert

Self-supervised

- Technical problem-solving and academic content authoring using MATLAB, CAD, and Python.

**Knowledge Cultivator**

Nov '23 – Feb '24

## PROJECTS

**Vision-Guided Robotic Arm for Simulated Oral Cancer Screening**

Aug '25 – Dec '25

**Autonomous Drone Navigation | MathWorks Global Student Drone Challenge 2025**

Mar '25 – Apr '25

**Air Quality Index Forecasting Using Time Series Models** [🔗](#)

Nov '24 – Nov '24

**3D Lunar Surface Modeling from DEM Data | ISRO Immersion Challenge 2024** [🔗](#)

Jul '24 – Jul '24

**Cricket Team Optimization | AMEX Campus Challenge 2024 (Product Track)** [🔗](#)

Jun '24 – Jul '24

**ExoMy Rover Navigation and UR3 Arm Motion Planning | ERC 2023 Remote** [🔗](#)

Apr '23 – Sep '23

## TEACHING EXPERIENCE

### Teaching Assistant

CS F320: Foundations of Data Science, BITS Pilani

Jan '26 – May '26

ME F218: Advanced Mechanics of Solids, BITS Pilani

Jan '25 – May '25

ME F216: Materials Science and Engineering, BITS Pilani

Sep '24 – Dec '24

## HONOURS & SCHOLARSHIPS

**Institute Merit-cum-Need Scholarship — Consecutive Recipient, BITS Pilani**

Oct '22 – May '26

## AWARDS & ACHIEVEMENTS

**Finalist — Meta PyTorch OpenEnv Hackathon (India)** [🔗](#)

Apr '26

**3<sup>rd</sup> Place — MathWorks Global Drone Student Challenge 2025**

Mar '25

**Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation)** [🔗](#)

Jul '24

**Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024** [🔗](#)

Jul '24

**5<sup>th</sup> Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)**

Sep '23

## LEADERSHIP EXPERIENCE

### Deputy Director of Academic Programming

Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania

Aug '26 – Present

### Vice-President

South Asian Association (Rangoli), University of Pennsylvania

Aug '26 – Present

### President & Secretary

Mechanical Engineering Association (MEA), BITS Pilani

Jun '24 – May '26

### President & Tech Fest Coordinator

Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani

Oct '24 – Jul '25

### Project Manager

Tinkerer's Lab (TL), BITS Pilani

May '24 – Jul '25

### Treasurer

Bengali Cultural Association (Moruchhaya), BITS Pilani

May '24 – Jul '25

## VOLUNTEERING EXPERIENCE

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### **Volunteer**

*Junoon, National Service Scheme (NSS), BITS Pilani*

*Sep '23 – Sep '23*

## PROFESSIONAL ACTIVITIES

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### **Graduate Student Member — IEEE**

*2026*

## COMPETITIVE EXAMINATIONS

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**GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)**

*2025*

**TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)**

*2025*

**JEE Advanced: All India Rank 14372**

*2022*

## TECHNICAL SKILLS & COURSEWORK

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**Relevant Coursework:** Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins

**Programming Languages:** Python, C++, C, C#, Java, Bash (Shell)

**Robotics & Simulation:** ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS

**AI & Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel

**Cloud & MLOps:** AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions

**Hardware & Embedded Systems:** NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR

**CAD & Mechanical Simulation:** ANSYS Mechanical, SolidWorks, Fusion 360